Importance Weighted Transfer Learning For High-dimensional Generalized Linear Regression

Zejing Zheng

Joint work with Junlong Zhao, Shengbin Zheng

School of Statistics, Beijing Normal University

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Background: High-Dimensional Challenge

Modeling high-dimensional data with small sample size is challenging:

1. Data Sparsity (Hall et al.,2005,JRSSB.)
2. Traditional methods Breakdown ($p > n$)
3. Statistical Instability

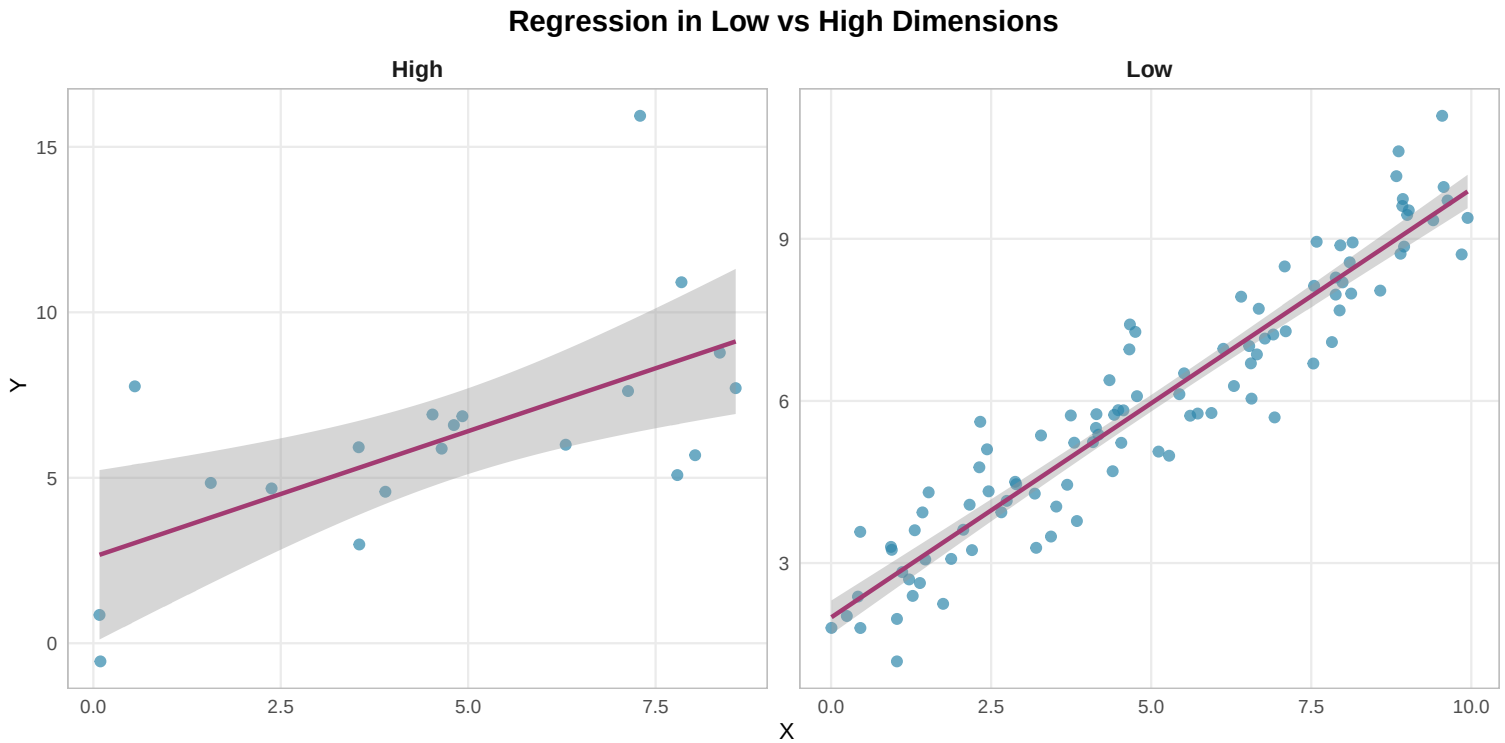
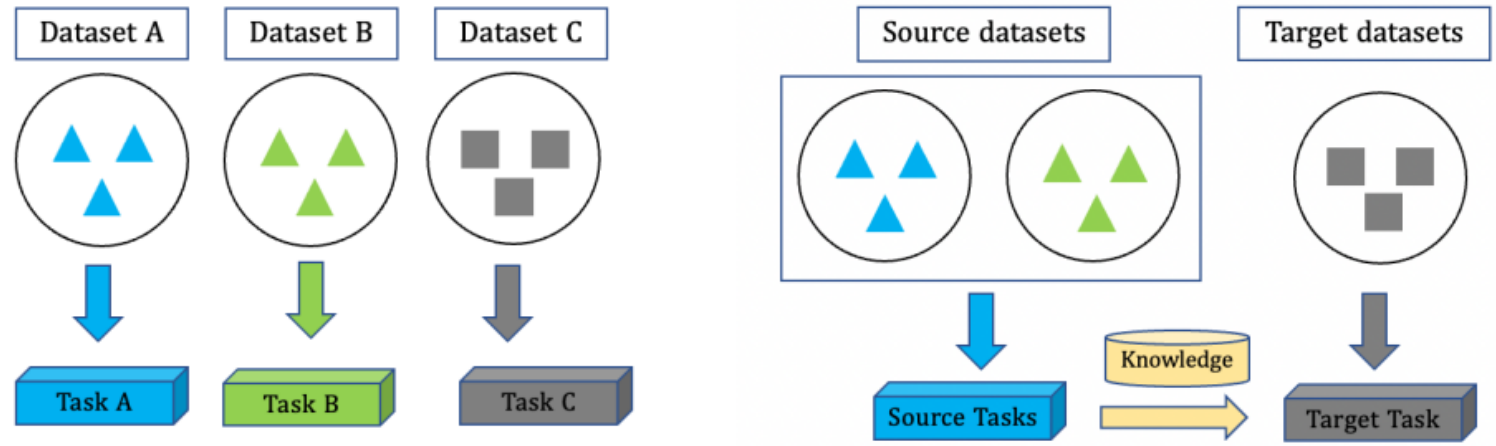


Figure 1

Background: transfer learning

- Classical methods use data solely on the target population;
- Transfer learning exploit source data to enhance the statistical inference on the target problem (Torrey and Shavlik, 2010; Weiss et al., 2016).

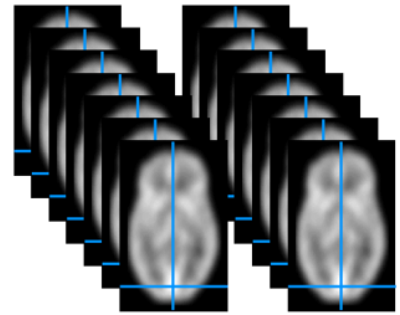


(a) Traditional Machine Learning (b) Transfer Learning

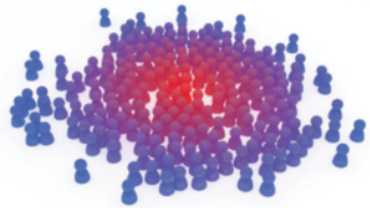
Figure 2: Processes between classical machine learning and transfer learning.

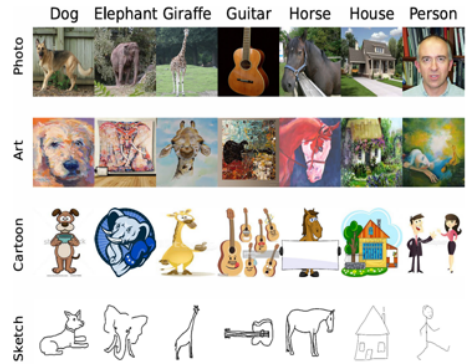
Applications

- In medical studies, rare diseases typically have small sample sizes, but transfer learning can enhance diagnostic accuracy via knowledge transfer across hospitals and diseases (Shakya et al., 2025).
- Pretrain a prediction model on COVID-19 data from high-data-volume countries, then transfer the model to predict case numbers in early-pandemic countries (Almutaani et al. 2024).
- In CV, pre-trained models (e.g., ResNet) learn general visual features (edges, textures) from massive general images (e.g., ImageNet). These features can be transferred to specific visual tasks, greatly reducing annotated data demand for target tasks (Ge et al. 2025).



statistical models in EPIDEMIOLOGY





Data and Heterogeneity types

Denote by $\boldsymbol{z} = (\boldsymbol{x}^\top, y)^\top \sim P^{(k)}(\boldsymbol{z})$ for $k = 0, 1$. There are three popular types of heterogeneities:

- Covariate shift: $P^{(0)}_{\boldsymbol{x}} \neq P^{(1)}_{\boldsymbol{x}}$, $P^{(0)}_{y|\boldsymbol{x}} = P^{(1)}_{y|\boldsymbol{x}}$ (Maity et al. 2021, Biometrika)
- Posterior drift: $P^{(0)}_{\boldsymbol{x}} = P^{(1)}_{\boldsymbol{x}}$, $P^{(0)}_{y|\boldsymbol{x}} \neq P^{(1)}_{y|\boldsymbol{x}}$ (Fang et al. 2020, NeurIPS)
- Label shift: $P^{(0)}_{y} \neq P^{(1)}_{y}$, $P^{(0)}_{\boldsymbol{x}|y} = P^{(1)}_{\boldsymbol{x}|y}$ (Maity et al. 2023, JMLR)
- Full-distribution shift: $P^{(0)}_{y} \neq P^{(1)}_{y}$, $P^{(0)}_{\boldsymbol{x}|y} \neq P^{(1)}_{\boldsymbol{x}|y}$ (Li et al. 2021, JRSSB, etc.).

Literature review

Transfer learning has been applied to high dimensional data modeling.

- High-dimensional linear models (Bastani,2021; Li et al. 2021, etc).
- High-dimensional GLM: Trans-GLM (Tian and Feng,2023), TransHDGLM (Li et al. 2023)
- Quantile regression (Zhang and Zhu,2022, etc.);
- Functional data analysis, gaussian graphical models, nonparametric regression and classification, etc.(Hanneke and Kpotufe, 2019; Cai and Wei, 2021; Reeve et al., 2021; Hanneke and Kpotufe, 2022; Gu et al., 2022; Fan et al., 2023; Xiong et al., 2023; Cai et al., 2024, etc.)

Data structure

For $k = 1, \cdots, K$, the observed data consist of:

- **Target Data**:

$$\mathcal{D}_0 = \left\{ \left((\boldsymbol{x}_i^{(0)})^\top, y_i^{(0)} \right)^\top, \ i = 1, \cdots, n_0 \right\} \overset{\text{i.i.d.}}{\sim} P_0$$

  with density function f_0

- **Source Data**:

$$\mathcal{D}_k = \left\{ \left((\boldsymbol{x}_i^{(k)})^\top, y_i^{(k)} \right)^\top, \ i = 1, \cdots, n_k \right\} \overset{\text{i.i.d.}}{\sim} P_k$$

  with density function f_k for $k = 1, \cdots, K$

- $n_k \ll p$ for $k = 0, \cdots, K$
- $\mathbb{E}(\boldsymbol{x}^{(k)}) = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{x}^{(k)}) = \boldsymbol{\Sigma}^{(k)}$ for $k = 0, \cdots, K$

High-dimensional Generalized Linear Models

GLM Framework

For $k = 0, \cdots, K$, assume $y^{(k)} \mid \boldsymbol{x}^{(k)}$ follows a GLM:

$$f(y^{(k)} \mid \boldsymbol{x}^{(k)}; \boldsymbol{\beta}^{(k)}) = \rho(y^{(k)}) \exp\left\{ y^{(k)} (\boldsymbol{x}^{(k)})^\top \boldsymbol{\beta}^{(k)} - \psi\left((\boldsymbol{x}^{(k)})^\top \boldsymbol{\beta}^{(k)} \right) \right\}$$

- For discrete $y^{(k)}$, f (density) is replaced by $\mathbb{P}$ (probability measure).
- $\boldsymbol{\beta}^{(k)} \in \mathbb{R}^p$ is an unknown sparse vector with $\|\boldsymbol{\beta}^{(k)}\|_0 = s_k \ll p$
- $\rho(\cdot)$ and $\psi(\cdot)$ are some known univariate functions

Inverse Link Function (McCullagh and Nelder, 1989)

$$\mathbb{E}(y^{(k)} \mid \boldsymbol{x}^{(k)}) = \psi'\left((\boldsymbol{x}^{(k)})^\top \boldsymbol{\beta}^{(k)} \right)$$

- $\psi'(\cdot)$ serves as the inverse link function.

High-dimensional GLM

- Typical examples:
  - Linear model with Gaussian noise: $\psi(u) = u^2/2$;
  - Logistic regression: $\psi(u) = \log[1 + \exp(u)]$;
  - Poisson regression: $\psi(u) = \exp(u)$.
- Goal: Improve the estimation of target parameter $\boldsymbol{\beta}^{(0)}$ by leveraging both the target and source data.

Condition 1

ψ is twice-differentiable and strictly convex.

Condition 2

Assume that $\max_{k,i} \|\boldsymbol{x}_i^{(k)}\|_\infty \leq U$ *and* $\sup_{|z| \leq \bar{U}} \psi''((\boldsymbol{x}_i^{(k)})^\top \boldsymbol{\beta}^{(k)} + z) \leq \Gamma_\psi$ *for all* i *and* k*, with some positive constants* Γ_ψ, U *and* $\bar{U}$.

Classical methods and limitations

- For $k = 0, \cdots, K$, denote the negative log-likelihood function as

$$\ell_{n,k}(y^{(k)}, \boldsymbol{x}^{(k)}; \boldsymbol{\beta}^{(k)}) = \frac{1}{n_k} \sum_{i=1}^{n_k} \left\{ -y_i^{(k)} (\boldsymbol{x}_i^{(k)})^\top \boldsymbol{\beta}^{(k)} + \psi((\boldsymbol{x}_i^{(k)})^\top \boldsymbol{\beta}^{(k)}) \right\}.$$

- Solve the following optimization problem with LASSO penalty on the target data only:

$$\hat{\boldsymbol{\beta}}_{\mathrm{Tar}}^{(0)} = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ \ell_{n,0}(y^{(0)}, \boldsymbol{x}^{(0)}; \boldsymbol{\beta}) + \lambda \|\boldsymbol{\beta}\|_1 \right\},$$

  where λ denotes the penalty parameter.

- (Limitations) The performance may be not satisfactory due to the limited target sample size n_0.

Two-step Trans-GLM method

Define the level-h informative set: $\mathcal{A} = \{k : \|\boldsymbol{\beta}^{(k)} - \boldsymbol{\beta}^{(0)}\|_1 \leq h\}$

- Step 1: Compute

$$\hat{\boldsymbol{\omega}} = \arg\min_{\boldsymbol{\omega}\in\mathbb{R}^p} \left\{ \frac{1}{n_0 + n_{\mathcal{A}}} \sum_{k\in\{0\}\cup\mathcal{A}} \ell_{n,k}(y^{(k)}, \boldsymbol{x}^{(k)}; \boldsymbol{\omega}) + \lambda_1 \|\boldsymbol{\omega}\|_1 \right\}$$

- Step 2: The resulting estimator is $\hat{\boldsymbol{\beta}}^{(0)} = \hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\delta}}$, where $\hat{\boldsymbol{\delta}}$ is defined as

$$\hat{\boldsymbol{\delta}} = \arg\min_{\boldsymbol{\delta}\in\mathbb{R}^p} \left\{ \frac{1}{n_0} \ell_{n,0}(y^{(0)}, \boldsymbol{x}^{(0)}; \hat{\boldsymbol{\omega}} + \boldsymbol{\delta}) + \lambda_2 \|\boldsymbol{\delta}\|_1 \right\}$$

Limitations The all-in-or-all-out way needs to identify the informative sources to avoid negative transfer, while ignoring the information hidden in the discarded sources.

Classical Importance Weighting

Importance Weighting Approach

$$\boldsymbol{\beta}^{(0)} = \arg\min \mathbb{E}_{(\boldsymbol{x},y)\sim f_0}[\ell(\boldsymbol{x}, y; \boldsymbol{\beta})] = \arg\min \mathbb{E}_{(\boldsymbol{x},y)\sim f_k}\left[\frac{f_0(\boldsymbol{x}, y)}{f_k(\boldsymbol{x}, y)}\ell(\boldsymbol{x}, y; \boldsymbol{\beta})\right]$$

- Importance weighting provides an alternative to avoid the negative transfer, which is developed as a classical sampling methods in statistical computing (Cochran, 1977; Fishman, 2013).
- Importance weighting achieves distribution matching by assigning weights to observations.
- **High-dimensional Challenge:**
  - Requires density ratio $f(\boldsymbol{x}, y)/g(\boldsymbol{x}, y)$ for high-dimensional $(\boldsymbol{x}, y)$
  - Suffers from curse of dimensionality
  - Density ratio estimation becomes unstable

Importance weighting in high-dimensional GLM

Proposition 1

For $k = 1, ..., K$, it holds that

$$\boldsymbol{\beta}^{(0)} = \underset{\boldsymbol{\beta} \in \mathbb{R}^p}{\operatorname{argmin}} \, \mathbb{E}_{\boldsymbol{z}^{(k)} \sim P_k}[\omega^{(k)} \cdot \ell_k(y^{(k)}, \boldsymbol{x}^{(k)}; \boldsymbol{\beta})] \tag{1}$$

where

$$\omega^{(k)} = \frac{f(y^{(k)}|\boldsymbol{x}^{(k)}; \boldsymbol{\beta}^{(0)})}{f_k(y^{(k)}|\boldsymbol{x}^{(k)}; \boldsymbol{\beta}^{(k)})} \tag{2}$$

represents the importance weight with $f(y|\boldsymbol{x}; \boldsymbol{\beta}^{(0)})$ standing for a generic conditional density function of y given $\boldsymbol{x}$ with the parameter $\boldsymbol{\beta}^{(0)}$, satisfying $\mathbb{E}_{y|\boldsymbol{x} \sim f}(y|\boldsymbol{x}; \boldsymbol{\beta}^{(0)}) = \psi'(\boldsymbol{x}^\top \boldsymbol{\beta}^{(0)})$. Here $\mathbb{E}_{y|\boldsymbol{x} \sim f}(y|\boldsymbol{x}; \boldsymbol{\beta}^{(0)})$ denotes the conditional expectation of y given $\boldsymbol{x}$ with $y|\boldsymbol{x} \sim f(y|\boldsymbol{x}; \boldsymbol{\beta}^{(0)})$.

- We first take $f(y|\boldsymbol{x};\boldsymbol{\beta}^{(0)})$ being $f_0(y|\boldsymbol{x};\boldsymbol{\beta}^{(0)})$ and assume that $y^{(k)}$ is bounded. The case of $y^{(k)}$ being unbounded will be addressed later.
- **Weights should be both upper and lower bounded.**

$$\omega_i^{(k)} = \frac{f_0(y_i^{(k)} \mid \boldsymbol{x}_i^{(k)};\boldsymbol{\beta}^{(0)})}{f_k(y_i^{(k)} \mid \boldsymbol{x}_i^{(k)};\boldsymbol{\beta}^{(k)})} = \frac{\exp\{y_i^{(k)}(\boldsymbol{x}_i^{(k)})^\top\boldsymbol{\beta}^{(0)} - \psi((\boldsymbol{x}_i^{(k)})^\top\boldsymbol{\beta}^{(0)})\}}{\exp\{y_i^{(k)}(\boldsymbol{x}_i^{(k)})^\top\boldsymbol{\beta}^{(k)} - \psi((\boldsymbol{x}_i^{(k)})^\top\boldsymbol{\beta}^{(k)})\}}$$

$$\begin{aligned} &= \exp\{-y_i^{(k)}\eta_i^{(k)} - \psi((\boldsymbol{x}_i^{(k)})^\top\boldsymbol{\beta}^{(0)}) + \psi((\boldsymbol{x}_i^{(k)})^\top\boldsymbol{\beta}^{(k)})\} \\ &= \exp\{-\eta_i^{(k)}\epsilon_i^{(k)0} - \psi''(\xi_i^{(k)})(\eta_i^{(k)})^2/2\} \end{aligned} \tag{3}$$

- $\eta_i^{(k)} = (\boldsymbol{x}_i^{(k)})^\top(\boldsymbol{\beta}^{(k)} - \boldsymbol{\beta}^{(0)})$
- $\epsilon_i^{(k)0} = y_i^{(k)} - \psi'((\boldsymbol{x}_i^{(k)})^\top\boldsymbol{\beta}^{(0)})$
- $\xi_i^{(k)} \in [(\boldsymbol{x}_i^{(k)})^\top\boldsymbol{\beta}^{(k)} - \eta_i^{(k)}, (\boldsymbol{x}_i^{(k)})^\top\boldsymbol{\beta}^{(k)}]$

Sample Selection for Bounded Responses

- For bounded $y^{(k)}$, the error term $\epsilon_i^{(k)0}$ is bounded
- Under Condition 1, $\psi''(\xi_i^{(k)})$ is bounded
- Controlling η_i's ensures stability

To obtain stable weights, we select observations from:

$$\mathcal{S}_k^b = \left\{ i \in [n_k] : |\eta_i^{(k)}| \leq M_k \right\}, \quad k = 1, \ldots, K$$

where

- $\eta_i^{(k)} = (\boldsymbol{x}_i^{(k)})^\top (\boldsymbol{\beta}^{(k)} - \boldsymbol{\beta}^{(0)})$
- M_k: tuning parameter selected via cross-validation

Sample Selection for Unbounded Responses

For unbounded $y^{(k)}$ (e.g., Poisson regression), $\epsilon_i^{(k)0}$ may be unbounded.

We choose samples in the set

$$\mathcal{S}_k^{un} = \left\{ i \in [n_k] : |\eta_i^{(k)}| \leq M_k,\ |\epsilon_i^{(k)0}| \leq A \right\}$$

equivalently,

$$\mathcal{S}_k^{un} = \left\{ i \in [n_k] : |\eta_i^{(k)}| \leq M_k,\ \left| y_i^{(k)} - \psi' \left((\boldsymbol{x}_i^{(k)})^\top \boldsymbol{\beta}^{(0)} \right) \right| \leq A \right\}$$

where A is a positive constant.

In summary, for $k = 1, \cdots, K$, we denote the selected subsets as

$$\mathcal{S}_k = \begin{cases} \mathcal{S}_k^b, & y^{(k)} \text{ is bounded;} \\ \mathcal{S}_k^{un}, & y^{(k)} \text{ is unbounded.} \end{cases}$$

Adjusted Weights for Unbounded $y^{(k)}$

When $y^{(k)}$ is **unbounded**, sample selection bias arise. We define the conditional density $f(y_i^{(k)} \mid \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}^{(0)})$ for unbounded $y^{(k)}$ as:

$$
\begin{aligned}
f^{\text{un}}(y_i^{(k)} \mid \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}^{(0)}) &= \frac{1}{2}\left[f_0(y_i^{(k)} \mid \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}^{(0)}) + f_0\left(-y_i^{(k)} + 2\psi'\left((\boldsymbol{x}_i^{(k)})^\top \boldsymbol{\beta}^{(0)}\right) \mid \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}^{(0)}\right)\right] \\
&= \frac{1}{2}\left[\tilde{f}_0(\epsilon_i^{(k)0} \mid \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}^{(0)}) + \tilde{f}_0(-\epsilon_i^{(k)0} \mid \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}^{(0)})\right]
\end{aligned}
$$

- This is the **symmetrized version** of the conditional density of the residual:

$$
\epsilon_i^{(k)0} = y_i^{(k)} - \psi'\left((\boldsymbol{x}_i^{(k)})^\top \boldsymbol{\beta}^{(0)}\right)
$$

In summary, the adjusted weights for $i = 1, \cdots, n_k$ and $k = 1, \cdots, K$ are defined as a piecewise function:

$$
\omega_i^{(k)} = \begin{cases} f_0(y_i^{(k)} \mid \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}^{(0)}) / f_k(y_i^{(k)} \mid \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}^{(k)}), & \text{if } y^{(k)} \text{ is bounded;} \\ f^{\text{un}}(y_i^{(k)} \mid \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}^{(0)}) / f_k(y_i^{(k)} \mid \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}^{(k)}), & \text{if } y^{(k)} \text{ is unbounded.} \end{cases}
$$

The oracle estimator

Proposition 2

Suppose that Conditions 1 and 2 are satisfied. For $k = 1, \ldots, K$ and all $i \in \mathcal{S}_k$, it holds that $b^l \leq \omega_i^{(k)} \leq b^u$ for some constants $0 < b^l < 1 < b^u < \infty$.

Assuming weights and the subsets $\mathcal{S}_k$'s are known, the oracle estimator is defined as

$$\tilde{\boldsymbol{\beta}}_{\text{ora}}^{(0)} = \underset{\boldsymbol{\beta} \in \mathbb{R}^p}{\operatorname{argmin}} \frac{1}{n_0 + n_{\text{sou}}} \left\{ \sum_{k=0}^{K} \sum_{i=1}^{n_k} \mathrm{I}\{i \in \mathcal{S}_k\} \omega_i^{(k)} \ell_k(y_i^{(k)}, \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}) \right\} + \lambda \|\boldsymbol{\beta}\|_1, \quad (4)$$

where $n_{\text{sou}} = \sum_{k=1}^{K} n_k$ denotes the sum of source sample size and λ is a tuning parameter.

Effective sample size

- With notation $\mathcal{S} = \cup_{k=1}^{K} \mathcal{S}_k$, write $n_{\mathcal{S}} = \sum_{k=1}^{K} n_{\mathcal{S}_k}$, where $n_{\mathcal{S}_k} = |\mathcal{S}_k|$.
- Sample usage rate (SUR) is defined as

$$\rho_{\mathcal{S}} = \frac{n_0 + \mathbb{E}(n_{\mathcal{S}})}{n_0 + \Sigma_{k=1}^{K} n_k}, \tag{5}$$

- Denote $\boldsymbol{\delta}^{(k)} = \boldsymbol{\beta}^{(k)} - \boldsymbol{\beta}^{(0)}$ and $d_k = \|\boldsymbol{\delta}^{(k)}\|_2$.
- When $\eta^{(k)} \sim \mathcal{N}(0, \kappa d_k^2)$ for some $\kappa \in [\kappa_l, \kappa_u]$, which holds when $\boldsymbol{x}_i^{(k)}$ follows a normal distribution, then

$$\mathbb{E}(n_{\mathcal{S}_k}) \asymp n_k \min\{M_k / d_k, 1\}. \tag{6}$$

The same conclusion holds if the density function of $\eta^{(k)}$ is lower and upper bounded respectively by the normal density functions with zero means in the region $(-M_k, M_k)$.

The properties of oracle IWTL-GLM

Condition 3

For any $\boldsymbol{u} \in \mathbb{R}^p$ such that $\|\boldsymbol{u}\|_2 \leq \delta$ with $\delta > 0$, there exist positive constants κ_1 and κ_2 such that

$$\frac{1}{n_0 + n_{\mathcal{S}}} \sum_{k=0}^{K} \sum_{i \in \mathcal{S}_k} ((\boldsymbol{x}_i^{(k)})^\top \boldsymbol{u})^2 \geq \kappa_1 \|\boldsymbol{u}\|_2 \left(\|\boldsymbol{u}\|_2 - \kappa_2 \sqrt{\frac{\log p}{n_0 + n_{\mathcal{S}}}} \|\boldsymbol{u}\|_1 \right),$$

with probability tending to one as $n_0 + n_{\mathcal{S}} \to \infty$.

Theorem 1

Suppose that Conditions 1-3 are satisfied. Let $\lambda \asymp \rho_{\mathcal{S}} \sqrt{\log p / (n_0 + \mathbb{E}(n_{\mathcal{S}}))}$, where $\rho_{\mathcal{S}} = (n_0 + \mathbb{E}(n_{\mathcal{S}}))/(n_0 + n_{\text{sou}})$. As $\min_{0 \leq k \leq K} n_k \to \infty$, it follows that

$$\|\tilde{\boldsymbol{\beta}}_{\text{ora}}^{(0)} - \boldsymbol{\beta}^{(0)}\|_2^2 = O_p \left(\frac{s_0 \log p}{n_0 + \mathbb{E}(n_{\mathcal{S}})} \right).$$

Estimation of IWTL-GLM

- The weights $\omega_i^{(k)}$'s and $\mathcal{S}_k$'s are unknown in practice.
- By plugging in the initial estimator $\tilde{\boldsymbol{\beta}}^{(k)}$'s (obtained from k-th source data only) into $\omega_i^{(k)}$ and $\mathcal{S}_k$, we can obtain the estimated weights $\hat{\omega}_i^{(k)}$ and selected subsets $\hat{\mathcal{S}}_k$.
- IWTL-GLM estimator, defined as follows:

$$\hat{\boldsymbol{\beta}}^{(0)} = \underset{\boldsymbol{\beta}\in\mathbb{R}^p}{\operatorname{argmin}} \frac{1}{n_0 + n_{\text{sou}}} \left\{ \sum_{k=0}^{K} \sum_{i=1}^{n_k} \mathrm{I}\{i \in \hat{\mathcal{S}}_k\} \hat{\omega}_i^{(k)} \ell_k(y_i^{(k)}, \boldsymbol{x}_i^{(k)}; \boldsymbol{\beta}) \right\} + \lambda \|\boldsymbol{\beta}\|_1.$$

Condition 4

For $k = 0, 1, \cdots, K$, the initial estimator $\tilde{\boldsymbol{\beta}}^{(k)}$ of $\boldsymbol{\beta}^{(k)}$ is independent of data in $\mathcal{S}_k$ and is consistent, satisfying $\|\tilde{\boldsymbol{\beta}}^{(k)} - \boldsymbol{\beta}^{(k)}\|_1 = O_p(\gamma_k)$ where $\gamma_k = o(1)$.

When $\tilde{\boldsymbol{\beta}}^{(k)}$ is constructed by the penalized likelihood method with LASSO penalty, it holds that $\gamma_k = s_k\sqrt{\log p / n_k}$.

Lemma 2

Suppose that Conditions 1, 2 and 4 are satisfied. For $1 \le k \le K$, as $\min_{0 \le k \le K} n_k \to \infty$, it holds that

$$\max_{i \in \hat{\mathcal{S}}_k} |\hat{\omega}_i^{(k)} / \omega_i^{(k)} - 1| = O_p(\gamma_0 \wedge \{\gamma_k + d_k + M_k\}). \tag{7}$$

Theorem 3

Suppose Conditions 1-4 are satisfied. Denote $\gamma_{\max} = \max_{1\le k\le K} \gamma_k$, $d_{\text{ave}} = \sum_{k=1}^{K} \pi_k d_k$ *and* $M_{\text{ave}} = \sum_{k=1}^{K} \pi_k M_k$ *with* $\pi_k = \mathbb{E}(n_{\mathcal{S}_k})/\mathbb{E}(n_{\mathcal{S}})$. *As* $\min_{0\le k\le K} n_k \to \infty$, *it holds that*

$$\|\hat{\boldsymbol{\beta}}^{(0)} - \boldsymbol{\beta}^{(0)}\|_2^2 = O_p\left(\frac{s_0 \log p}{n_0 + \mathbb{E}\left(n_{\mathcal{S}}\right)} + s_0\{\gamma_0 \wedge (\gamma_{\max} + d_{\text{ave}} + M_{\text{ave}})\}^2\right). \quad (8)$$

- The error of our estimator is no worse than γ_0^2——the error using only target data, avoiding negative transfer without choosing informative set.
- If $\tilde{\beta}_0$ is LASSO estimator (i.e. $\gamma_0 = s_0\sqrt{\log p/n_0}$), when both d_{ave} and M_{ave} are less than γ_0, then IWTL-GLM estimator is better than LASSO.

Corollary 4 (Two sources)

Assume that (i) $n_1 = \cdots = n_K = N \gg n_0$ *with a finite constant* K*; (ii)* $M_k \equiv \min_{1 \le k \le K} d_k$ *for all* $k = 1, \ldots, K$*. Furthermore, assuming that there exist some* $k' \in [K]$ *such that* $d_{k'} \ll \sqrt{\log p / n_0}$*, it holds that* $d_{\text{ave}} = O(d_{k'}) \ll \sqrt{\log p / n_0}$*, and then and then*

$$\|\hat{\boldsymbol{\beta}}^{(0)} - \boldsymbol{\beta}^{(0)}\|_2^2 = O_p\left(\frac{s_0 \log p}{n_0 + n_{\text{sou}}} + s_0 d_1^2\right). \tag{9}$$

- This implies that, as long as there exist sources that are relatively close to the target, the IWTL-GLM estimator can achieve the positive transfer under mild conditions.

Simulation Results: Linear regression model

Competitors: RIW-TL (our method) and its variants, RIW-TL-P and RIW-TL-U, plus the Trans-Lasso method and the LASSO method using target data only.

Parameter settings:

- $n_0 = 150, p = 200, K = 10$ and $n_k = 600$ for $k = 1, \cdots, K$.
- **Covariates.** $\boldsymbol{x}_i^{(0)}$ is $i.i.d.$ from a multivariate normal distribution $\mathcal{N}_p(\mathbf{0}_p, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} = (0.5^{|j-j'|})$; $\mathbf{x}_i^{(k)}$ for $k = 1, \cdots, K$ are $i.i.d.$ from a central multivariate t distribution $t_p(\mathbf{0}_p, 5, \boldsymbol{\Sigma})$;
- **Response.** With the coefficients specified as later, the response data are generated by

$$y_i^{(k)} = (\boldsymbol{x}_i^{(k)})^\top \boldsymbol{\beta}^{(k)} + \epsilon_i^{(k)}, \ k = 0, 1, \cdots, K,$$

where the errors $\epsilon_i^{(k)} \sim N(0, 1)$.

Simulation Results: Linear regression model

Define an index set $\mathcal{B} = [m_{\mathcal{B}}]$ where $m_{\mathcal{B}} = 0, 2, 4, 6, 8$ or 10. Set $\boldsymbol{\beta}^{(0)} = (\mathbf{1}_{s_0}^\top, \mathbf{0}_{p-s_0}^\top)^\top$ with $s_0 = 10$. For $1 \le k \le K$, we specify $\boldsymbol{\beta}^{(k)}$ as follows.

(i) For a given $\mathcal{B}$, if $k \in \mathcal{B}$, let

$$\beta_j^{(k)} = \beta_j^{(0)} - 0.5\mathrm{I}(j \in T_k),$$

where T_k is a random subset of $\{s_0 + 1, \cdots, p\}$ with $|T_k| = d$. The values of d will be taken later in different simulations.

(ii) For a given $\mathcal{B}$, if $k \notin \mathcal{B}$, let

$$\beta_j^{(k)} = \begin{cases} \beta_j^{(0)} - 1, & j \in [s_0]; \\ \beta_j^{(0)} - 0.5, & j \in H_k; \\ \beta_j^{(0)}, & \text{otherwise}, \end{cases}$$

where H_k is a random subset of $\{s_0 + 1, \cdots, p\}$ with $|H_k| = 2s_0$.

Simulation Results: Linear regression model

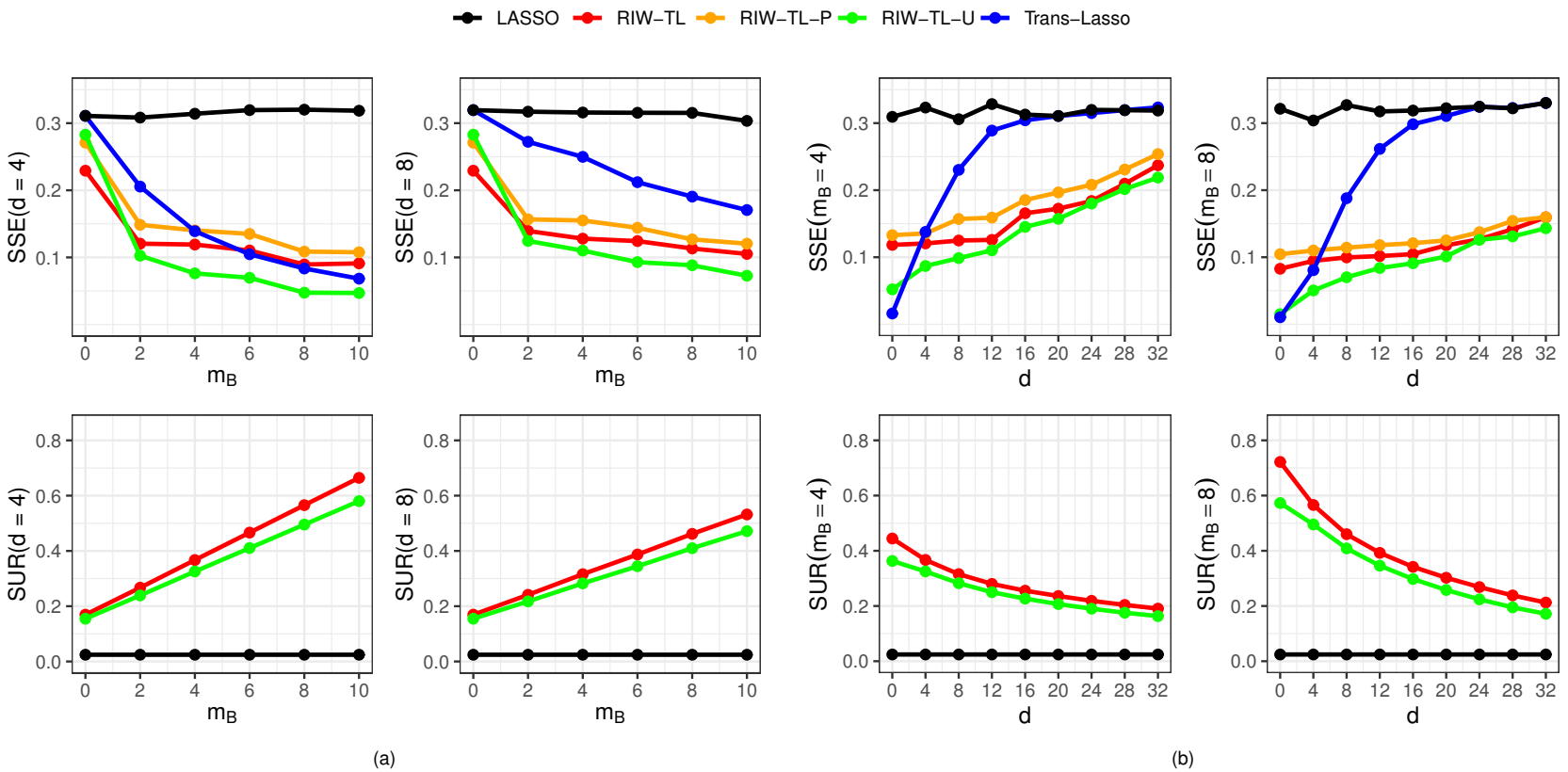


Figure 3: Full Distribution Shift: (a) Estimation errors (top row) and sample usage rates (bottom row) vs. m_B for different d values. (b) Estimation errors (top row) and sample usage rates (bottom row) vs. d for different m_B values. In the SUR plots, RIW-TL-P lines overlap with RIW-TL and are not visible..

Simulation Results: Logistic regression model

Competitors:

- LASSO: the penalized likelihood estimator with ℓ_1 penalty using only the target data
- IWTL-GLM (IWTL); aTransHDGLM(TR-HD); Trans-GLM(TR-GLM).

Parameter settings:

- $n_0 = 500$, $p = 500$, $n_k = 500$ for $k \in \{1, 2, ..., K\}$ with $K = 10$.
- **Covariates:** $\boldsymbol{x}_i^{(k)}$ for $k = 0, \cdots, K$ is $i.i.d.$ from a multivariate normal distribution $\mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} = (0.5^{|j-j'|})$; $\mathbf{x}_i^{(k)}$ for $k = 1, \cdots, K$ are $i.i.d.$ from a central multivariate t distribution $t_p(\mathbf{0}_p, 4, \boldsymbol{\Sigma})$
- **Response:** With the coefficients specified as later, the response data are generated from the logistic regression models.

Simulation Results: Logistic regression model

Set the target parameter $\boldsymbol{\beta}^{(0)} = (\mathbf{1}_{s_0}^\top, \mathbf{0}_{p-s_0}^\top)^\top$, with $s_0 = 5$. For $1 \le k \le K$, we specify $\boldsymbol{\beta}^{(k)}$ as follows.

(i) For $k = 1, \cdots, K_0$, let

$$\beta_j^{(k)} = \beta_j^{(0)} - 0.5\mathrm{I}(j \in G_k),$$

where G_k is a random subset of $\{s_0 + 1, \cdots, p\}$ with $|G_k| = d$. The values of d will be taken later in different simulations.

(ii) For $k = K_0 + 1, \cdots, K$, let

$$\beta_j^{(k)} = \begin{cases} \beta_j^{(0)} - 1, & j \in [s_0]; \\ \beta_j^{(0)} + 1, & j \in H_k; \\ \beta_j^{(0)}, & \text{otherwise}, \end{cases}$$

where G_k is a random subset of $\{s_0 + 1, \cdots, p\}$ with $|G_k| = 2s_0$.

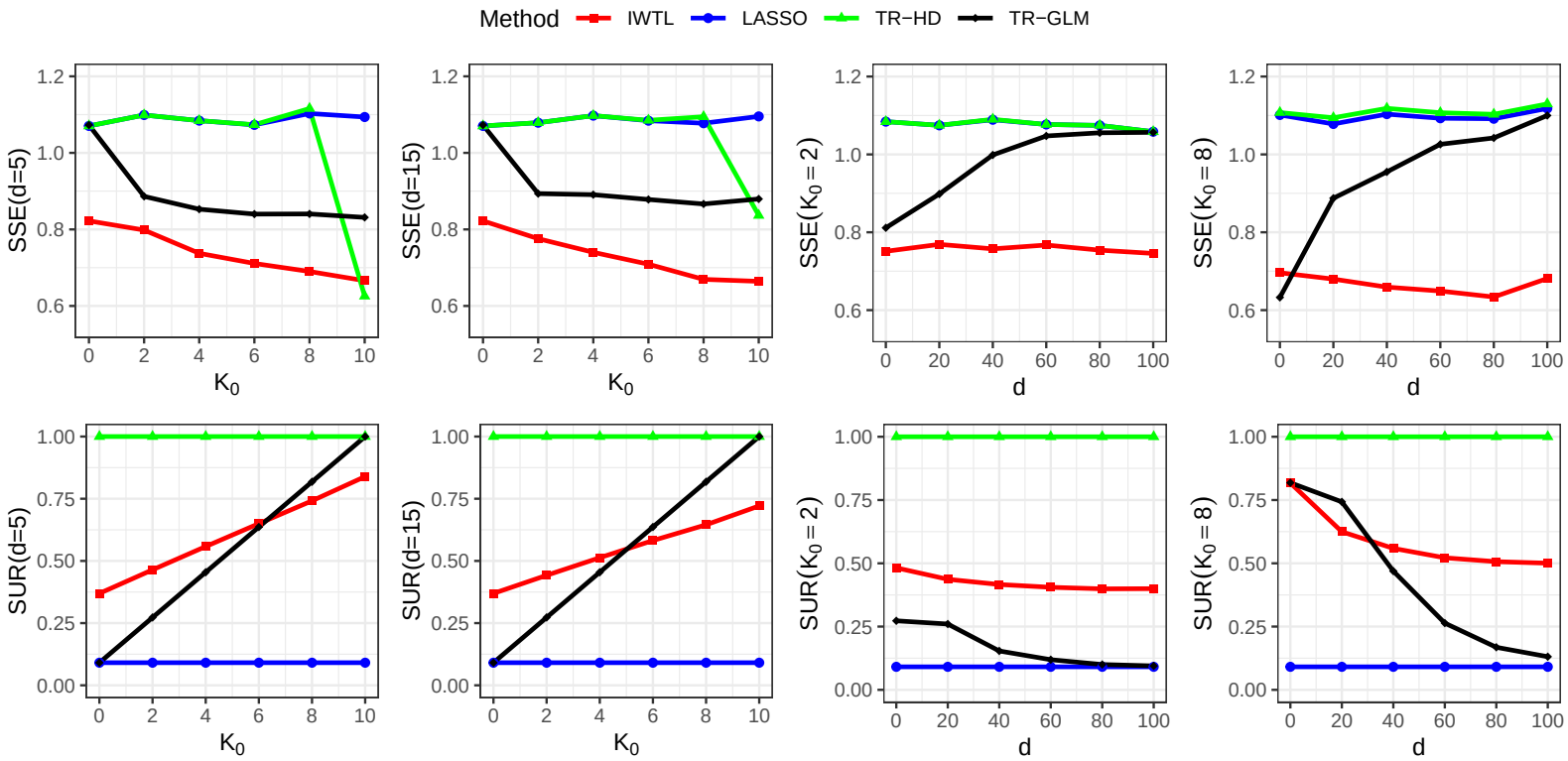


Figure 4: The results for logistic regression model under full distribution shift: (i) the first two columns show the SURs and SSEs over K_0 for different d; (ii) the last two columns show the SURs and SSEs over d for different K_0.

Real data analysis

Apartment rental data set:

- It includes 13 Beijing districts (e.g., Daxing and Fengtai) with sample size ranging from 200 to 350. Each district is treated as the target in turn while the others districts as sources.
- Y: the desirability of apartments, which is recorded as 1 if it exceeds the district average and 0 otherwise;
- $\boldsymbol{X}$: there are $p = 22$ predictors, including the number of rooms, the total number of floors within the building and other 20 concerning the condition of apartment;
- Goal: utilize these predictors to assess the desirability of an apartment within its district, which is conducive on pricing strategy assistance and resource optimization allocation;
- Based on the logistic regression model, we compare our method IWTL-DR (IWTL) with the three competitors, i.e., LASSO, Trans-GLM (TR-GLM) and aTransHDGLM (TR-HD).

Real data analysis

Metrics:

- The average classification error rates (AER) are reported over 100 replications. The smaller value of AER implies the superior performance.
- Sample Usage Rates (SUR): Blue denotes the target district, white represents regions excluded from the dataset, and varying color shades of other districts indicate SUR magnitude (darker = higher SUR).

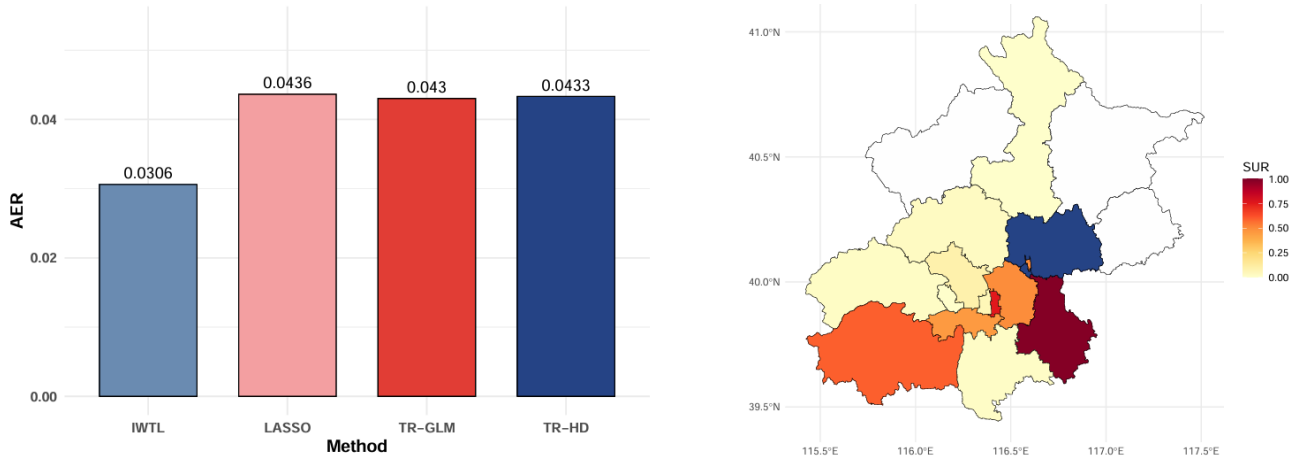


Figure 5: The districts “Shunyi” is taken as the target: (i) the left panel is about the results of AER for different methods; (ii) the right panel shows the SURs of different districts in IWTL-DR.

Real data analysis

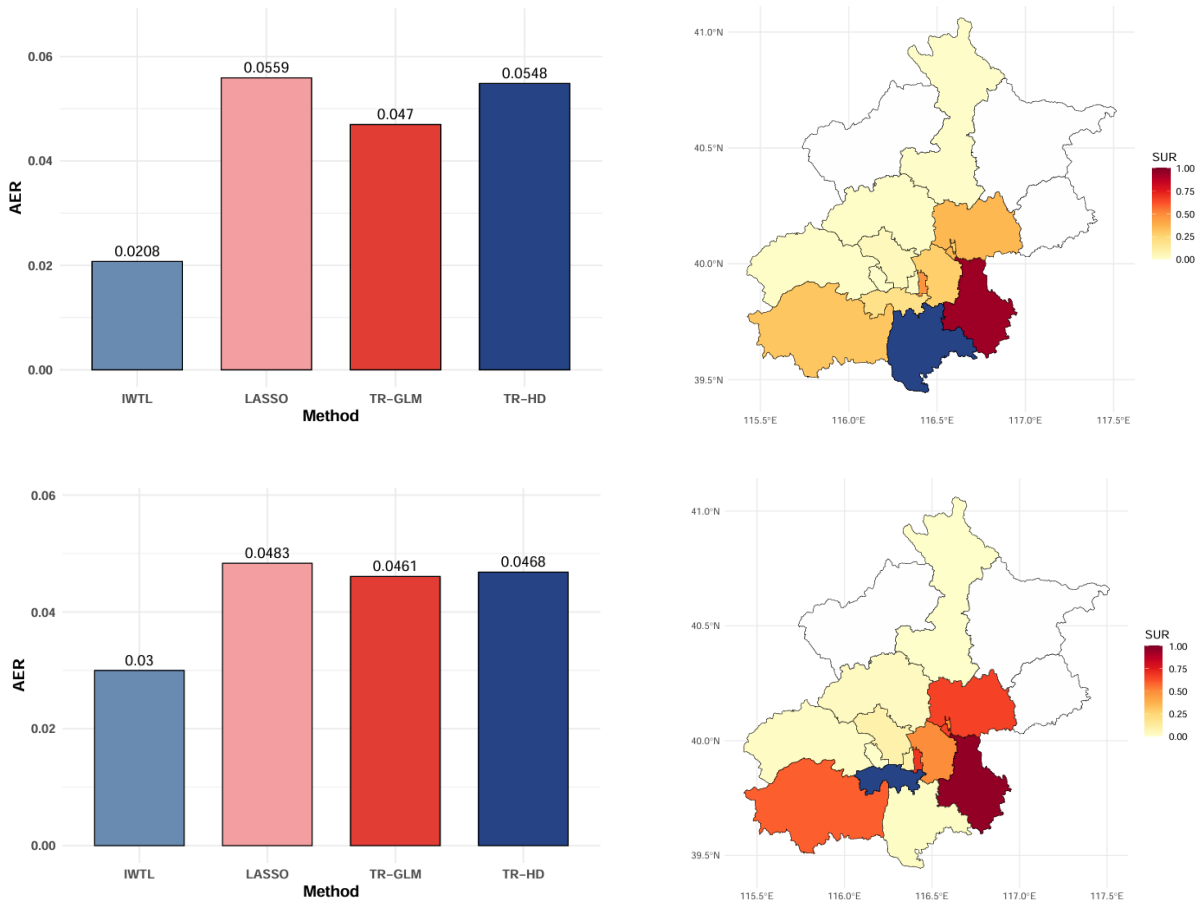


Figure 6: The districts “Daxing” (the top) and “Fengtai” (the bottom) are taken as the targets: (i) the left column is about the results of AER for different methods; (ii) the right column shows the SURs of different districts in IWTL-DR.

Summary

- We propose an importance weighted transfer learning method for high dimensional GLM (IWTL-GLM), where the weights are constructed from the one-dimensional density ratio, avoiding the cure of dimensionality.

$$\boldsymbol{\beta}^{(0)} = \underset{\boldsymbol{\beta}\in\mathbb{R}^p}{\operatorname{argmin}}\, \mathbb{E}_{\boldsymbol{z}^{(k)}\sim f_k}\left[\frac{f(y^{(k)}|\boldsymbol{x}^{(k)};\boldsymbol{\beta}^{(0)})}{f_k(y^{(k)}|\boldsymbol{x}^{(k)};\boldsymbol{\beta}^{(k)})}\cdot \ell_k(y^{(k)},\mathbf{x}^{(k)};\boldsymbol{\beta})\right].$$

- Different from the existing methods that select source data in all-in-or-all-out manner, our method assign different weights to observations individually, making more efficient use of source data.
- Our method can avoid negative transfer without selecting the informative sources.
- The idea can be extended to robust regression and semi-parametric regression, etc.

References

1. Zhao, J.$^{\#}$, S. Zheng$^{\#}$, and C. Leng* (2023). Residual importance weighted transfer learning for high-dimensional linear regression. arXiv:2311.07972.
2. Zheng, Z.$^{\#}$, S. Zheng$^{\#}$, and J. Zhao* (2025). Transfer Learning For High Dimensional Data With Discrete Responses. Computational Statistics & Data Analysis.

Thanks for your attention!